Pooja Prasad

AI Engineer | Full-Stack Developer | MSc in CS

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SUMMARY

AI Engineer with an MSc in Computer Science (Bauhaus-Universität Weimar) and **3 years** of software engineering experience, including internships, at **Bosch, Juniper Networks and Mindtree**. Experienced in building **REST APIs, backend services and frontends** (Angular, React, Node JS) and working directly with clients. Developed and evaluated **LLM and transformer-based NLP systems**: designed a custom **annotation** platform with **GPT-5**-assisted labeling, fine-tuned **BERT-based transformer models** (PyTorch, Hugging Face) for **spam detection**, trained an **AI-text detector with 99.5% accuracy** applied to **8M+ web pages**, and ran **batch inference** on an HPC cluster with **Docker and SLURM**. Team player with strong communication in both German and English.

TECHNICAL SKILLS

Machine Learning and NLP: PyTorch, Hugging Face Transformers, scikit-learn, supervised learning, feature engineering, model evaluation, prompt engineering, Large Language Models (LLMs), Generative AI, RAG, LangGraph, Agentic AI, TensorFlow, PySpark

Data and Evaluation: Pandas, NumPy, Tableau, Matplotlib, Seaborn, statistical analysis (Pearson correlation), temporal trend analysis, annotation design, embeddings, text classification, web data extraction (Resiliparse)

MLOps and Cloud: Docker, Kubernetes, Google cloud platform, SLURM (HPC), distributed batch inference, CI/CD pipeline, Jenkins, Git version control, JIRA

Programming Languages: Python, Java, TypeScript, JavaScript, SQL, HTML, CSS, PHP, C/C++

Backend and APIs: Spring Boot, microservices, System Architecture, Node JS, Express, REST APIs, Web security (JWT, OAuth, RBAC, OWASP), System Design

Frontend: React, Next.js, Angular, Redux, RxJS, jQuery, Tailwind CSS, SASS, Bootstrap

Databases: MySQL, MongoDB

Testing: Jest, Enzyme, Jasmine, Karma

DATA SCIENCE RESEARCH

Master's Thesis: A Web-Scale Analysis of Trends in AI-Generated and SEO Spam Content Using Human Annotation and ML
Bauhaus-Universität Weimar, Webis research group poojaprasadrn.github.io/msc-thesis-transformers

- Analyzed **8.07M search-result pages** from Bing, DuckDuckGo and Startpage (2022–2024) to track **AI-generated and spam content** over time.
- Built a **custom web annotation platform** and led a **9-annotator study** on 300 web pages, reaching **Krippendorff's alpha of 0.84**; the resulting labels served as ground truth for the spam classifier.
- Engineered a **two-stage review and spam classifier** on a 300-page labeled set, evaluating 8 classical and transformer-based configurations. A hybrid **DistilBERT** (embeddings plus linguistic features) performed best in both stages and was selected as the final model.
- Ran batch inference on ~360,000 pages across **4 HPC nodes** with **SLURM and Docker/Singularity**, cutting processing time by **~70%** vs. sequential runs.
- Trained a **TF-IDF + linear SVM AI-text detector** (**99.5% accuracy** on the PAN 2024 test set) and applied it to 8M+ pages at ~6 ms per page on CPU.
- Found that the **AI-generated share rose steadily** on all three engines yet stayed around 1% or below, and was **largely independent of spam**. **Spam rose significantly only on Startpage**, with no significant trend on Bing or DuckDuckGo.
- Built a **RAG (retrieval-augmented generation)** question-answering tool over my Master's thesis in Python: TF-IDF retrieval, a **LangGraph workflow** with query rewriting, relevance grading and a retry/clarify fallback, and LLM-generated answers with page citations.

PROFESSIONAL EXPERIENCE

Web Developer Intern (Angular) 10/2023 – 04/2024 | Stuttgart, Germany
Robert Bosch GmbH

- Built a **dashboard** for the technical sales team in **Angular and TypeScript** for Bosch's Origify project.
- Migrated legacy dashboards to a new **Angular** framework, backed by unit tests in **Jasmine and Karma**.
- Cut page load time by **~90%** by optimizing **RxJS-based state management**.

- Turned requirements into **UX improvements** and built new features.
- Collaborated with cross-functional teams in an **Agile/Scrum environment**: sprint planning, stand-ups and retrospectives.

Full-Stack Web Developer (MERN stack) 12/2020 – 09/2022
Mindtree Limited

- Completed **3-month MERN full-stack training** (MongoDB, Express, React, Node.js) before joining the project team.
- Built **responsive, cross-browser** core modules of a client web application in **React, Next.js, Redux and Tailwind CSS**; supported its production go-live.
- Reduced load times by **47%** by optimizing **Redux** state management. Worked with **Google Cloud Platform (GCP)** in the project's cloud environment.
- Raised unit test coverage by **52% with test-driven development (TDD), Jest and Enzyme** in a **Jenkins CI/CD pipeline**. Cut code review time by **21%** by enforcing **ESLint** standards.
- Ran knowledge-transfer sessions, mentored juniors and worked directly with clients in **Agile/Scrum** teams using **Git** across the full **Software Development Life Cycle.**

Software Engineering Intern 07/2019 – 08/2019
Juniper Networks

- Built a **Python tool** for memory-footprint analysis of JUNOS routers (MX/PTX) using **PyEZ and Matplotlib**.
- Correlated memory use with routing-table size to estimate the **memory cost per route** and modeled baseline fluctuation from **time-series data**, documented the tool for team reuse.
- Collected router telemetry with **Healthbot and InfluxDB** for anomaly monitoring and benchmarked network performance with iPerf.

Android Developer Intern 08/2019 – 03/2020
Bosch Limited

- Built an **Android app in Java** for bicycle rental and docking on large campuses.
- Set up the **Java and Spring Boot microservices** backend for the app.
- Developed **OTP-based authentication on Raspberry Pi** to secure bicycle access.

Full-Stack Developer Intern 07/2018 – 08/2018
Zool Tech Solutions

- Built a full-stack movie booking web application using **Laravel/CodeIgniter (PHP) and MySQL for backend** route handling, user authentication, and seat reservation logic, paired with an interactive **HTML/CSS/Bootstrap frontend**.

EDUCATION

Master of Science: Computer Science for Digital Media 10/2022 – 01/2026 | Germany
Bauhaus-Universität Weimar
Final grade 1.8 • Thesis grade 1.3

- **Master's Thesis: A Web-Scale Analysis of Trends in AI-Generated and SEO Spam Content Using Human Annotation and Machine Learning**

Bachelor of Engineering: Computer Science and Engineering 2016 – 2020
Global Academy of Technology
Graduated with Distinction

- **Bachelor's Thesis: Sustainable transportation solutions for large campuses – mobile application with a Java and Spring Boot microservices backend**

CERTIFICATIONS AND TRAINING

- **Core Java and Python**, IIHT (2019, grade A)
- **Big Data Analytics and Machine Learning** workshop, IIIT Allahabad (2019)
- **Secure Delivery for Web Developers** and **Node.js and React foundation** courses, Mindtree (2021)
- **Techradiance '18** (IIT Bombay): **2nd position**, zonal round at GAT, Bengaluru (2017)

LANGUAGES

English (C2 Level) • **German** (B1 Level)